

01 - Power Measurement IP

Build a system that can measure electrical parameters like voltage and current in real time and calculate power accurately for smart electrical monitoring applications.

SDG 7 - Affordable & Clean Energy

SDG 9 - Industry, Innovation & Infrastructure

Accurate real-time measurement of electrical parameters such as voltage, current, and power is essential for smart monitoring applications. Existing solutions often rely on bulky external modules, which limit scalability and integration in chip-level systems.

The Challenge

Design and implement a chip-level IP core that measures voltage and current in real time, performs RMS calculations, and outputs accurate power values. The solution must be lightweight, efficient, and suitable for integration into embedded SoCs for smart energy monitoring.

Mandatory Features

Each feature must be implemented as part of a unified solution:

1. Voltage & Current Sensing Interface

- Support sensor inputs for electrical signals.
- Ensure isolation and safety in measurement.
- Provide stable analog-to-digital conversion readiness.

2. ADC Data Acquisition System

- Read analog signals via ADC.
- Enable real-time sampling with minimal latency.
- Store readings with timestamps for verification.

3. RMS Calculation Module

- Implement hardware-efficient RMS algorithms.
- Maintain accuracy across varying loads and frequencies.
- Optimize for chip-level resource constraints.

4. Real-Time Power Measurement Output

- Compute instantaneous and average power.
- Provide digital outputs for integration with monitoring systems.
- Support interfacing with embedded dashboards or SoCs.

Bonus Features (Optional)

- Cloud-based analytics for multi-device monitoring.
- AI-driven anomaly detection in power usage.
- Integration with IoT platforms for predictive maintenance.

02 - Motor Control IP

Create a system that can control motor speed and operation efficiently, similar to systems used in electric vehicles and industrial automation.

SDG 7 - Affordable & Clean Energy

SDG 9 - Industry, Innovation & Infrastructure

Efficient motor control is critical in applications ranging from electric vehicles to industrial automation. Traditional systems often rely on external controllers, which limit scalability and integration. A chip-level IP solution can provide precise speed regulation, protection, and optimized performance directly within embedded systems.

The Challenge

Design and implement a chip-level IP core that generates PWM signals, controls motor speed and frequency, and integrates protection logic. The solution must ensure reliable operation, safeguard motors against faults, and provide real-time output control suitable for embedded SoCs.

Mandatory Features

Each feature must be implemented as part of a unified solution:

1. PWM Signal Generation Module

- Generate precise PWM signals for motor drive.
- Support variable duty cycles for speed control.
- Ensure stable frequency output.

2. Speed & Frequency Control System

- Adjust motor speed dynamically.
- Provide frequency modulation for efficiency.
- Log control parameters for monitoring.

3. Motor Protection Logic

- Detect overcurrent, overheating, or stall conditions.
- Implement automatic shutdown or safe mode.
- Ensure compliance with safety standards.

4. Motor Output Control

- Deliver controlled signals to motor drivers.
- Support multiple motor types (DC, BLDC).
- Provide real-time feedback for closed-loop control.

Bonus Features (Optional)

- AI-driven optimization of motor efficiency.
- Cloud-based monitoring of motor health.
- Integration with IoT platforms for predictive maintenance

03 - Video Watermarking IP

Build a system that can insert a hidden watermark inside video frames to help detect tampering and protect digital content.

SDG 9 - Industry, Innovation & Infrastructure

SDG 16 - Peace, Justice & Strong Institutions

Digital content is increasingly vulnerable to tampering, piracy, and unauthorized distribution. Traditional watermarking methods are often software-based, making them easier to bypass and less efficient for real-time applications. A chip-level IP solution can embed hidden watermarks directly into video frames, ensuring robust protection and authenticity.

The Challenge

Design and implement a chip-level IP core that reads video frames, inserts hidden watermarks, and processes frames in real time. The solution must ensure that the watermark is imperceptible to viewers but detectable during verification, thereby safeguarding digital content integrity.

Mandatory Features

Each feature must be implemented as part of a unified solution:

1. Video Frame Reader

- Capture and process video frames in real time.
- Support multiple video formats and resolutions.
- Ensure minimal latency during frame acquisition.

2. Watermark Insertion Module

- Embed hidden watermark data into video frames.
- Ensure imperceptibility to human viewers.
- Provide configurable watermark strength and patterns.

3. Real-Time Frame Processing

- Maintain high throughput for continuous video streams.
- Optimize for FPGA/ASIC resource constraints.
- Guarantee synchronization between watermark insertion and frame output.

4. Protected Video Output

- Deliver watermarked video streams without quality loss.
- Support verification of watermark presence.
- Enable integration with digital rights management systems.

Bonus Features (Optional)

- AI-driven watermark pattern adaptation for enhanced security.
- Cloud-based verification of watermarked content.
- Multi-level watermarking for layered protection.

04 - Audio DSP IP

Design a system that can process audio signals in real time, such as filtering noise and improving sound quality.

SDG 9 - Industry, Innovation & Infrastructure

SDG 12 - Responsible Consumption & Production

High-quality audio processing is essential in applications such as communication systems, entertainment devices, and hearing aids. Traditional software-based solutions often introduce latency and consume excessive resources. A chip-level IP core can deliver real-time audio filtering, gain control, and output processing with efficiency and precision.

The Challenge

Design and implement a chip-level IP core that processes audio signals in real time, reduces noise, and enhances sound quality. The solution must integrate digital filters, gain control, and audio output modules while maintaining low latency and resource efficiency.

Mandatory Features

Each feature must be implemented as part of a unified solution:

1. Audio Input System

- Capture audio signals from microphones or line inputs.
- Support multiple sampling rates.
- Ensure minimal distortion during acquisition.

2. Digital Filter Implementation

- Apply noise reduction and signal enhancement filters.
- Support configurable filter parameters.
- Optimize for real-time performance on FPGA/ASIC.

3. Gain Control Module

- Adjust audio levels dynamically.
- Prevent clipping and distortion.
- Provide automatic gain control (AGC) capability.

4. Audio Output Processing

- Deliver processed audio to speakers or external systems.
- Maintain synchronization with input streams.
- Ensure high-quality, low-latency output.

Bonus Features (Optional)

- AI-driven adaptive filtering for dynamic environments.
- Multi-channel audio support (stereo/surround).
- Cloud-based audio analytics for monitoring and optimization.

05 - Edge Analytics IP

Create a lightweight system that can analyze sensor data instantly on edge devices without depending on cloud processing.

SDG 9 - Industry, Innovation & Infrastructure

SDG 11 - Sustainable Cities & Communities

Modern IoT systems generate massive amounts of sensor data, but relying solely on cloud processing introduces latency, bandwidth costs, and privacy concerns. A chip-level IP core for edge analytics can enable instant, lightweight data processing directly on devices, improving responsiveness and efficiency.

The Challenge

Design and implement a chip-level IP core that collects sensor data, applies real-time filtering, and generates analytics outputs without cloud dependency. The solution must be lightweight, efficient, and suitable for integration into embedded IoT systems.

Mandatory Features

Each feature must be implemented as part of a unified solution:

1. Sensor Data Collection System

- Interface with multiple sensor inputs.
- Ensure reliable acquisition with minimal noise.
- Provide timestamped data for processing.

2. Moving Average Filter

- Implement a hardware-efficient moving average algorithm.
- Smooth sensor data to reduce fluctuations.
- Support configurable window sizes.

3. Real-Time Data Processing

- Perform analytics directly on edge devices.
- Maintain low latency and minimal resource usage.
- Optimize for embedded system constraints.

4. Output Analytics System

- Generate processed insights for immediate use.
- Provide digital outputs for dashboards or controllers.
- Support integration with IoT platforms.

Bonus Features (Optional)

- AI-driven anomaly detection at the edge.
- Multi-sensor fusion for advanced analytics.
- Cloud sync option for hybrid edge-cloud systems.

06 - BOM-Optimized SOM (System on Module)

Design a modular hardware system that reduces overall cost while maintaining good performance for industrial applications.

SDG 9 - Industry, Innovation & Infrastructure

SDG 12 - Responsible Consumption & Production

Industrial applications often require modular hardware systems that balance performance with cost efficiency. Traditional designs can be expensive due to complex BOM (Bill of Materials) requirements, limiting scalability and adoption. A BOM-optimized SOM can reduce costs while maintaining reliable performance.

The Challenge

Design and implement a modular hardware system-on-module (SOM) that minimizes BOM costs without compromising performance. The solution must provide a flexible architecture, expandable interfaces, and cost optimization strategies suitable for industrial applications.

Mandatory Features

Each feature must be implemented as part of a unified solution:

1. Basic Peripheral Modules

- Include essential peripherals (I/O, communication, storage).
- Ensure compatibility with industrial standards.
- Provide modular plug-and-play functionality.

2. Modular System Architecture

- Design a scalable SOM structure.
- Support integration of multiple modules.
- Optimize for reusability across applications.

3. Expandable Interface Design

- Provide interfaces for additional modules.
- Support flexible connectivity (SPI, I2C, UART, etc.).
- Ensure easy expansion without redesign.

4. Cost Optimization Strategy

- Minimize BOM costs through efficient design.
- Use cost-effective components without performance loss.
- Document trade-offs between cost and efficiency.

Bonus Features (Optional)

- AI-driven BOM optimization suggestions.
- Cloud-based monitoring of SOM performance.
- Support for rapid prototyping and reconfiguration.

07 - Video Pipeline Core

Develop a system that receives raw video input, processes image frames, and produces synchronized video output in real time.

SDG 9 - Industry, Innovation & Infrastructure

SDG 11 - Sustainable Cities & Communities

Modern applications such as surveillance, broadcasting, and multimedia systems require efficient video pipelines that can process raw input, synchronize frames, and deliver real-time output. Traditional software-based solutions often introduce latency and resource overhead, making them unsuitable for high-performance embedded systems.

The Challenge

Design and implement a chip-level IP core that receives raw video input, synchronizes frames, applies image processing, and produces real-time video output. The solution must ensure low latency, high throughput, and seamless integration into FPGA/ASIC-based systems.

Mandatory Features

Each feature must be implemented as part of a unified solution:

1. Video Input Handling System

- Capture raw video streams from sensors or external sources.
- Support multiple resolutions and formats.
- Ensure minimal latency during acquisition.

2. Frame Synchronization Module

- Align and synchronize video frames in real time.
- Prevent frame drops or mismatches.
- Provide timestamped synchronization logs.

3. Image Processing Block

- Apply basic image enhancement or filtering.
- Optimize for hardware resource constraints.
- Maintain real-time performance across varying workloads.

4. Video Streaming Output

- Deliver processed video streams with stable quality.
- Support integration with display or transmission systems.
- Ensure synchronization between input and output pipelines.

Bonus Features (Optional)

- AI-driven video enhancement (noise reduction, super-resolution).
- Multi-stream support for parallel video channels.
- Cloud-based video analytics integration.

08 - Multi-Camera Aggregator

Build a system that can combine and synchronize multiple camera inputs into a single organized output stream

SDG 9 - Industry, Innovation & Infrastructure

SDG 11 - Sustainable Cities & Communities

Applications such as surveillance, autonomous vehicles, and industrial monitoring often require multiple camera inputs to be combined and synchronized. Traditional systems struggle with latency, frame mismatches, and inefficient switching, limiting real-time performance. A chip-level IP core can aggregate multiple camera streams into a unified, synchronized output.

The Challenge

Design and implement a chip-level IP core that receives multiple camera inputs, synchronizes frames, manages input switching, and produces a combined video output stream. The solution must ensure low latency, high reliability, and seamless integration into FPGA/ASIC-based systems.

Mandatory Features

Each feature must be implemented as part of a unified solution:

1. Multi-Camera Input System

- Support multiple camera interfaces simultaneously.
- Handle varying resolutions and formats.
- Ensure reliable acquisition with minimal frame loss.

2. Frame Synchronization Logic

- Align frames from different cameras in real time.
- Prevent mismatches or dropped frames.
- Provide synchronization metadata for verification.

3. Input Switching Mechanism

- Enable dynamic switching between camera inputs.
- Support priority-based or event-driven switching.
- Maintain seamless transitions without glitches.

4. Combined Video Output

- Merge synchronized inputs into a unified stream.
- Support integration with display or transmission systems.
- Ensure stable, high-quality output.

Bonus Features (Optional)

- AI-driven scene selection from multiple cameras.
- Cloud-based multi-camera analytics.
- Support for panoramic or stitched video outputs.

09 - SDR (Software Defined Radio) Core

Create a communication system that can transmit and receive digital signals using software-controlled radio functions.

SDG 9 - Industry, Innovation & Infrastructure

SDG 17 - Partnerships for the Goals

Modern communication systems demand flexibility to support multiple standards and protocols. Traditional hardware radios are rigid, requiring separate designs for each modulation scheme. A chip-level SDR core enables software-controlled modulation and demodulation, providing adaptability, scalability, and efficiency for next-generation wireless systems.

The Challenge

Design and implement a chip-level IP core that transmits and receives digital signals using software-defined radio functions. The solution must include modulation, demodulation, synchronization, and error testing, ensuring reliable communication across diverse wireless environments.

Mandatory Features

Each feature must be implemented as part of a unified solution:

1. Signal Modulation Module

- Implement digital modulation schemes (e.g., QPSK, OFDM).
- Support configurable parameters for flexibility.
- Ensure efficient hardware resource usage.

2. Signal Demodulation Module

- Decode incoming signals accurately.
- Handle noise and interference effectively.
- Provide real-time demodulated data output.

3. Synchronization Logic

- Align transmitter and receiver timing.
- Prevent data loss due to drift or jitter.
- Maintain stable communication channels.

4. Error Testing Mechanism

- Implement error detection and correction (CRC, FEC).
- Provide performance metrics (BER, SNR).
- Log communication errors for analysis.

Bonus Features (Optional)

- AI-driven adaptive modulation based on channel conditions.
- Multi-band support for diverse wireless standards.
- Cloud-based monitoring of SDR performance.

10 - Quantization Engine

Build a system that reduces the size of AI models so they can run faster and efficiently on hardware with limited resources

SDG 9 - Industry, Innovation & Infrastructure

SDG 12 - Responsible Consumption & Production

AI models are often too large and computationally intensive to run efficiently on hardware with limited resources. This leads to high latency, increased power consumption, and restricted deployment in embedded systems. A chip-level quantization engine can reduce model size, accelerate inference, and enable efficient AI execution on constrained devices.

The Challenge

Design and implement a chip-level IP core that performs quantization of AI models, enabling faster and more resource-efficient inference. The solution must support INT8 and fixed-point computations, integrate compression logic, and optimize AI workloads for embedded hardware.

Mandatory Features

Each feature must be implemented as part of a unified solution:

1. INT8 Processing Module

- Support integer-based computations for AI inference.
- Ensure accuracy while reducing model size.
- Provide configurable precision levels.

2. Fixed-Point Computation Block

- Implement fixed-point arithmetic for efficient processing.
- Optimize for low-power embedded systems.
- Maintain stability across diverse workloads.

3. Data Compression Logic

- Reduce memory footprint of AI models.
- Implement hardware-efficient compression techniques.
- Ensure minimal accuracy loss during compression.

4. AI Inference Optimization

- Accelerate inference through quantized execution.
- Provide performance metrics (latency, throughput).
- Enable seamless integration with embedded AI frameworks.

Bonus Features (Optional)

- AI-driven dynamic quantization for adaptive workloads.
- Multi-precision support (INT4, INT16).